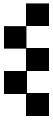
Characterization of IR System



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IR System Characterization

- IR system can be characterized at different levels based on
  - types of users
  - types of data
  - types of the information need
  - size and scale of the information repository

Types of Users

- Users can greatly vary in their abilities to interact with computational systems.
- The user may be
  - an expert user (for example, a curator or a librarian) who is searching for specific information that is clear in his/her mind, understands the scope and the structure of the available repository, and forms relevant queries for the task
  - a layperson user with a generic information need.
- The latter cannot create highly relevant queries for search (for example, students trying to find information about a new topic, or a person trying to shop for clothing).

Types of Data

- Search systems can be tailored to specific types of data.
- A vertical search engine is distinct from a general web search engine, in that it focuses on a specific segment of online content.
- **Common verticals** include shopping, the automotive industry, legal information, medical information, scholarly literature, job search and travel.
- **Examples of vertical search engines** include the Library of Congress, Mocavo, Nuroa, Trulia and Yelp.

Types of Information Need (1)

- In the context of Web search, users' information needs may be defined as
  - navigational,
  - informational, or
  - transactional

Types of Information Need (2)

- **Navigational search** refers to finding a particular piece of information (such as the Georgia Tech University Web site) that a user needs quickly.
- The purpose of **informational search** is to find current information about a topic (such as research activities in the college of computing at Georgia Tech—this is the classic IR system task).
- The goal of **transactional search** is to reach a site where further interaction happens resulting in some transactional event(such as joining a social network, shopping for products, and so on).

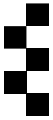
Levels of Scale (1)
Enterprise Search Systems

- Enterprise search systems offer IR solutions for searching different entities in an enterprise's intranet.
- The searchable entities include
  - e-mails,
  - corporate documents,
  - manuals,
  - charts,
  - and presentations,
  - as well as reports related to people, meetings, and projects.

Levels of Scale (2)
Desktop Search Engines

- Desktop search engines for retrieving files, folders, and different kinds of entities stored on the computer.
- **For Example,** Google Desktop and OS X Spotlight
- There are ***other systems that use peer-to-peer technology***, such as the BitTorrent protocol, which allows sharing of music in the form of audio files, **as well as specialized search engines for audio**, such as Lycos and Yahoo! audio search.

Modes of Interaction in IR System



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Modes of Interaction

- There are two main modes of interaction with IR systems
  - Retrieval
  - Browsing

- similar in goal
- accomplished through different interaction tasks

Terminology
Collection and Query

- The collection is made up of documents containing unstructured data.
- A query is a set of terms (also referred to as keywords) used by the searcher to specify an information need

For example

- The terms databases and operating systems
- query may also be a natural language phrase or a question (“What is the currency of China?” or “Find Italian restaurants in Florida.”).

Retrieval and Browsing

- **Retrieval** is concerned with the **extraction of relevant** information from a repository of documents through an IR query,
- **Browsing** signifies the **exploratory activity** of a user visiting or navigating through similar or related documents based on the user's assessment of relevance.
  - **Hyperlinks** are used to interconnect Web pages and are mainly used for browsing.
  - **Anchor** texts are text phrases within documents used to label hyperlinks and are very relevant to browsing.

Example of Browsing

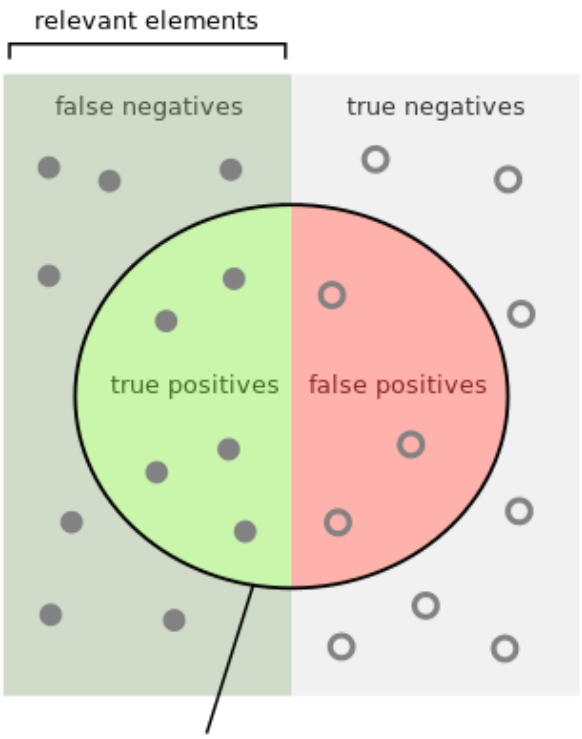
Article Talk Read Edit View history Search Wikipedia

Precision and recall

From Wikipedia, the free encyclopedia

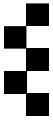
In pattern recognition, information retrieval and classification (machine learning), **precision** (also called positive predictive value) is the fraction of relevant instances among the retrieved instances, while **recall** (also known as sensitivity) is the fraction of the total amount of relevant instances that were actually retrieved. Both precision and recall are therefore based on an understanding and measure of relevance.

Suppose a computer program for recognizing dogs in photographs identifies 8 dogs in a picture containing 10 cats and 12 dogs (the relevant elements). Of the 8 identified as dogs, 5 actually are dogs (true positives), while the other 3 are cats (false positives). 7 dogs were missed (false negatives), and 7 cats were correctly excluded (true negatives). The program's precision is 5/8 (true positives / all positives) while its recall is 5/12 (true positives / relevant elements). When a search engine returns 30 pages, only 20 of which were relevant, while failing to return 40



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Search Engines And Web Crawlers



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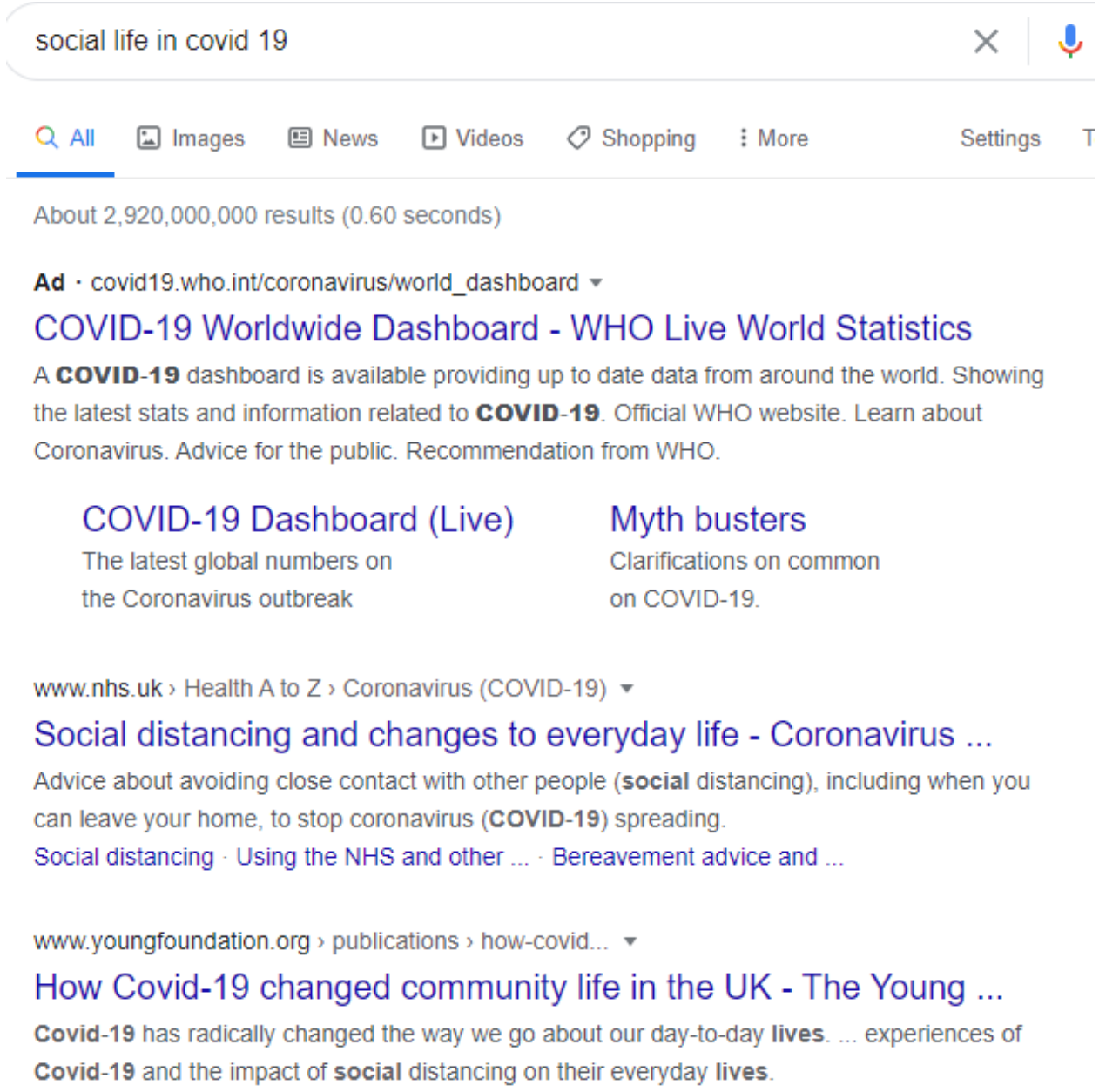
Search Engine

- A search engine is a **practical application of information retrieval** to large-scale document collections.
- People today have interactive access to enormous amounts of user-generated distributed content on the Web.
- This has spurred the rapid growth in search engine technology, where search engines are trying to discover different kinds of real-time content found on the Web.
- The part of a search engine responsible for discovering, analyzing, and indexing these new documents is known as a crawler.

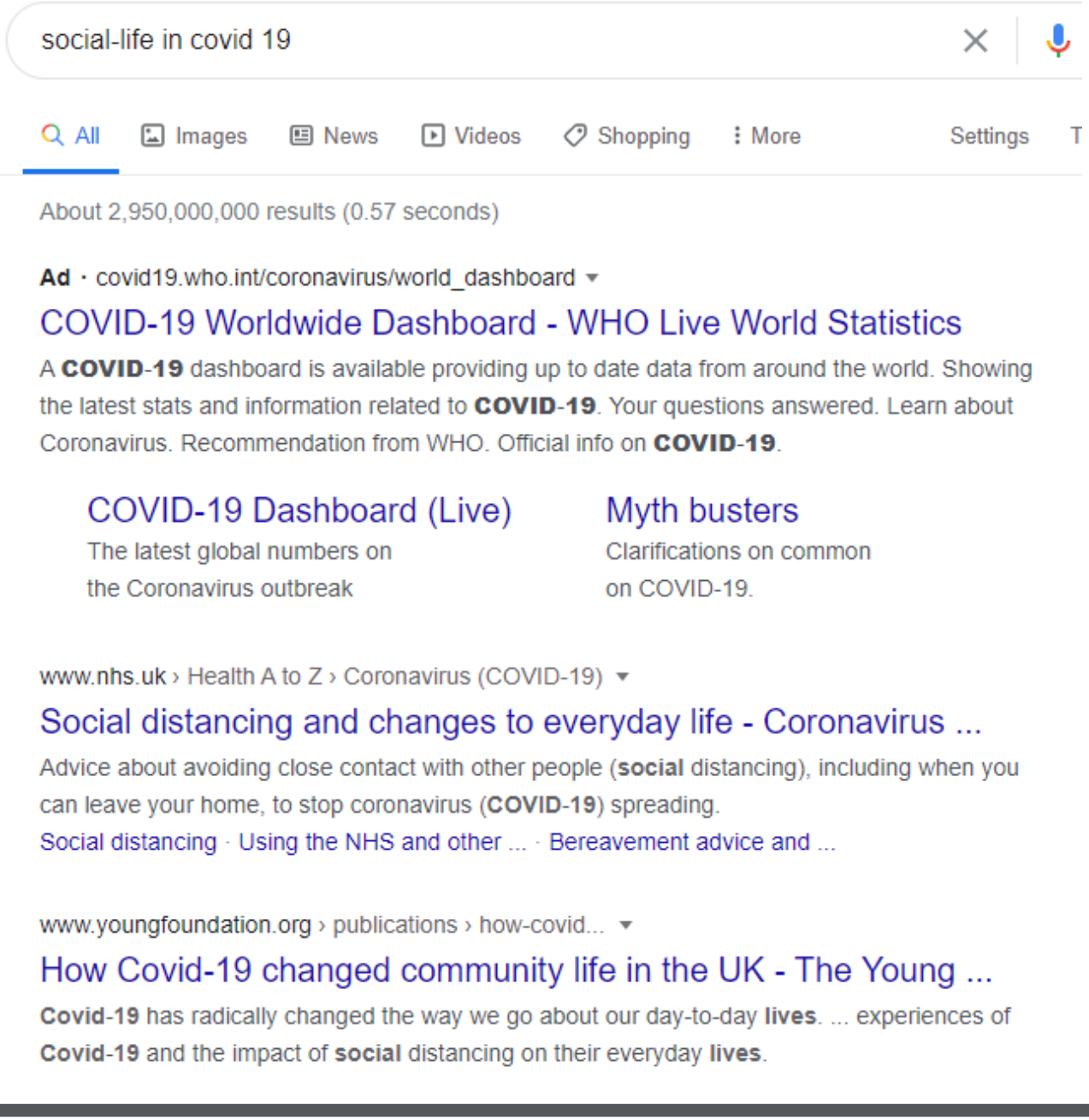
Domain-specific Search Engines

- Other types of search engines exist for specific domains of knowledge.
  - For example, the biomedical literature search database was started in the 1970s and is now supported by the PubMed search engine, which gives access to over 24 million abstracts.

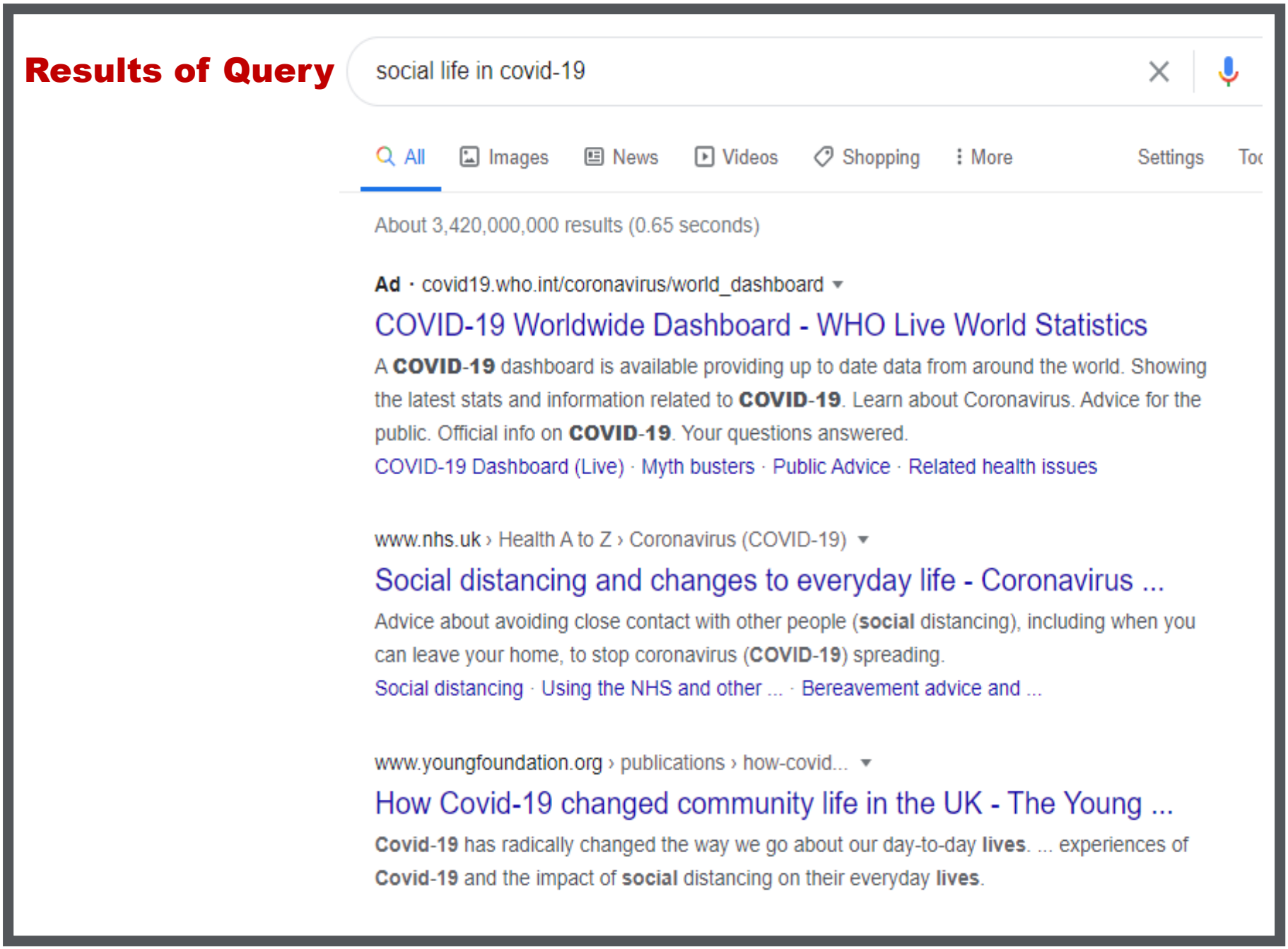
Results of Query



Results of Query



Results of Query



Results of Query

vida social en covid 19

All | Images | Videos | News | Maps | More | Settings

About 1,100,000,000 results (0.56 seconds)

Ad · covid19.who.int/coronavirus/world_dashboard

COVID-19 Worldwide Dashboard - WHO Live World Statistics

A **COVID-19** dashboard is available providing up to date data from around the world. Showing the latest stats and information related to **COVID-19**. Advice for the public. Recommendation from WHO. Learn about Coronavirus. Your questions answered.

COVID-19 Dashboard (Live)

The latest global numbers on the Coronavirus outbreak

Myth busters

Clarifications on common on COVID-19.

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La vida social del Covid-19: una etnografía del ... - U A B

La **vida social** del **Covid-19**: una etnografía del escepticismo y el negacionismo en poblaciones informales de Ciudad de México.

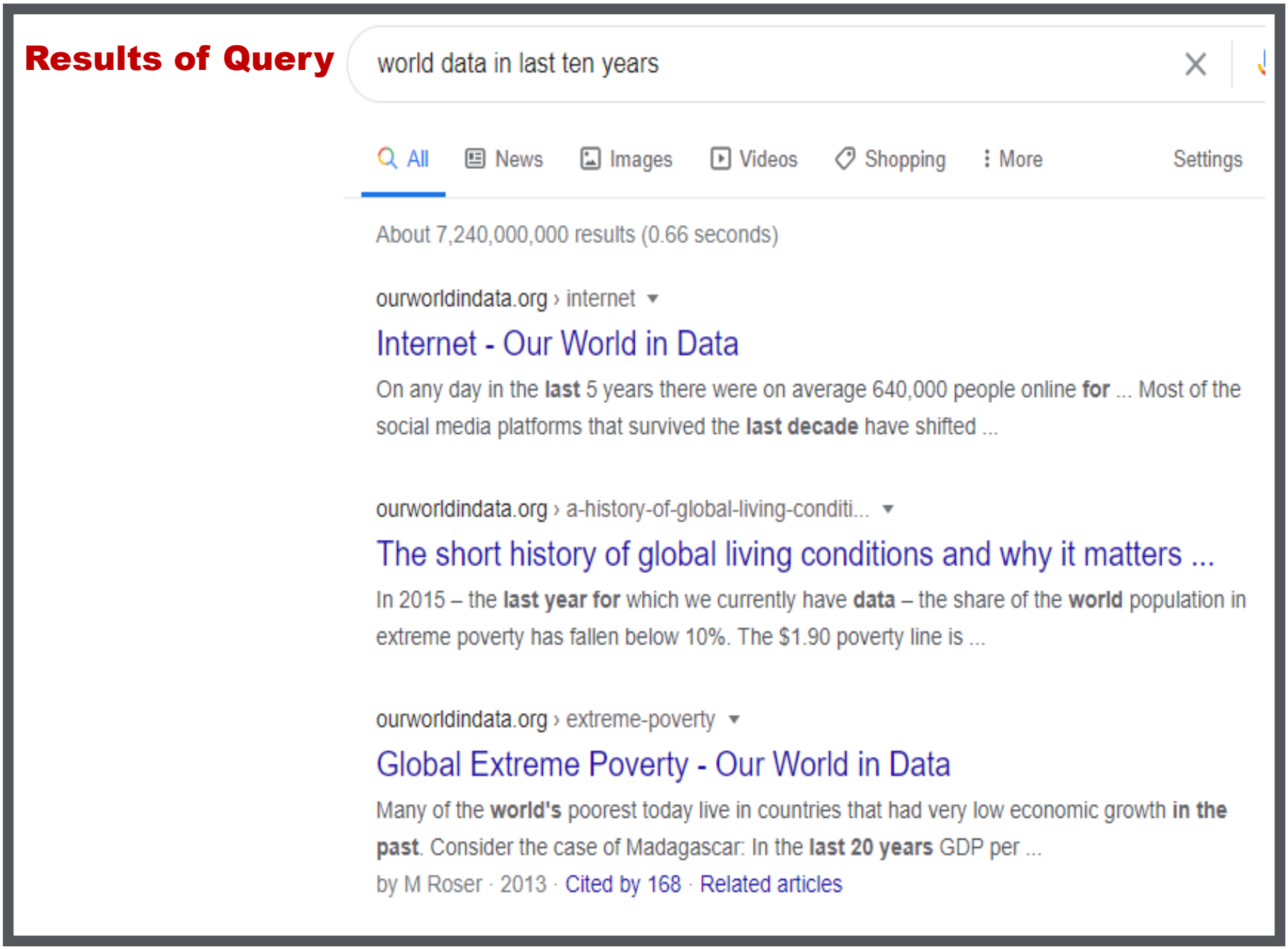
by JT Pérez · 2020

www.lifespan.org › infectious-diseases ▾ PDF Translate this page

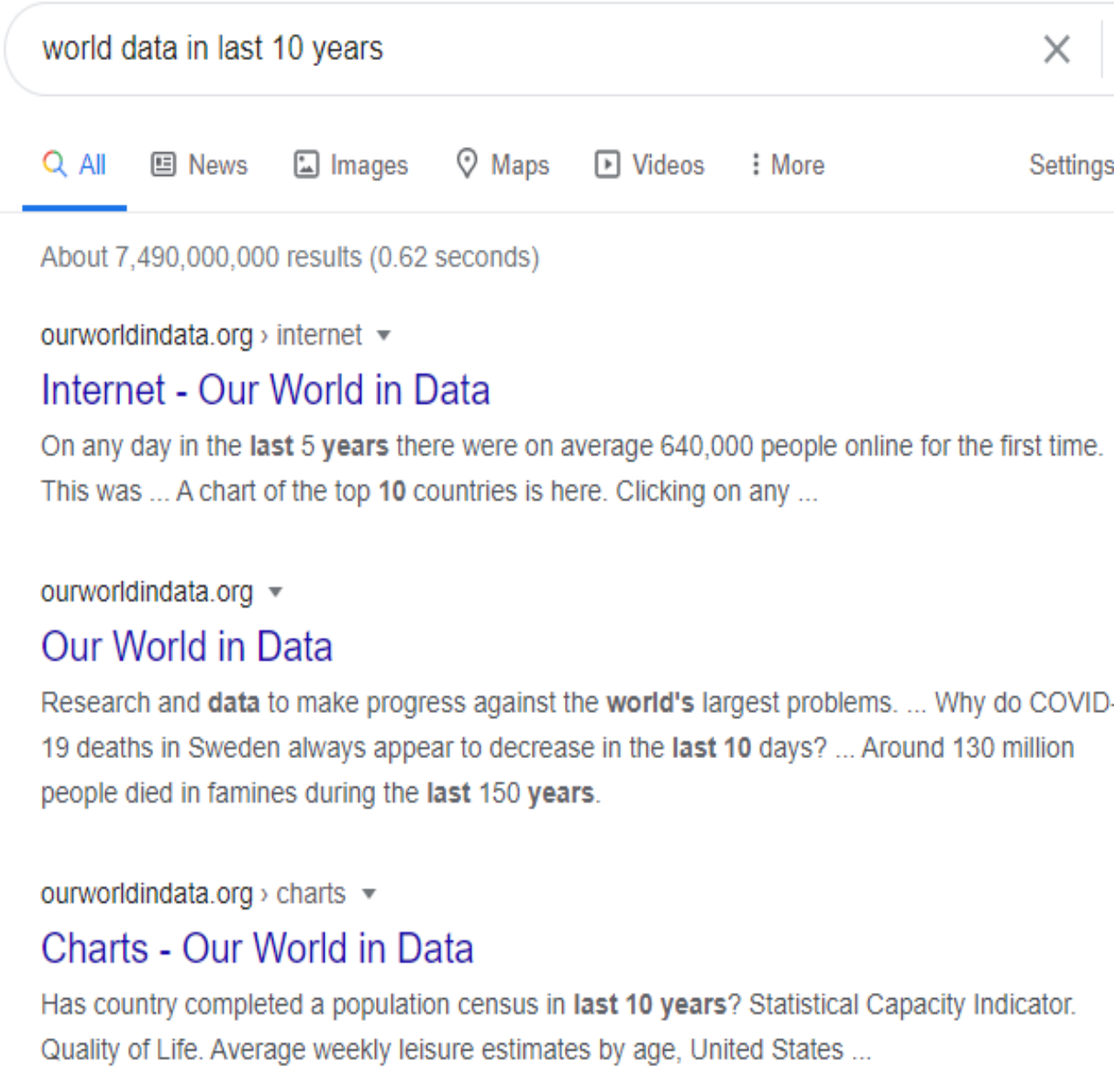
COVID-19 y cómo afecta nuestra vida cotidiana - Lifespan

En las últimas semanas, aprendimos nuevos términos como «distanciamiento **social**»,. «autoaislamiento» y «aplanamiento de la curva» para referirnos a

Results of Query



Results of Query



WWW and Google Indexing

How many indexed Web pages are there?

5.59 billion pages

The size of the World Wide **Web** (The **Internet**) The **Indexed Web** contains at least 5.59 billion **pages** (Wednesday, 11 November, 2020).

www.worldwidewebsize.com

WorldWideWebSize.com | The size of the World Wide Web (The ...

Indexing. After a page is discovered, **Google** tries to understand what the page is about. This process is called **indexing**. **Google** analyzes the content of the page, catalogs images and video files embedded on the page, and otherwise tries to understand the page.

support.google.com › webmasters › answer

How Google Search Works - Search Console Help

How long does it take for Google to index a site?

Although it varies, it seems to **take** as little as 4 days and up to 6 months for a **site** to be crawled by **Google** and attribute authority to the domain. When you publish a new blog post, **site page**, or website in general, there are many factors that determine how quickly it will be **indexed** by **Google**.

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More on Search Engines (1)

- Web search combines both aspects—browsing and retrieval.
- Web pages are like documents.
- Web search engines maintain an indexed repository of Web pages, usually using the technique of inverted indexing.

More on Search Engines (2)

- They retrieve the most relevant Web pages for the user in response to the user's search request with a possible ranking in descending order of relevance.
- The rank of a Web page in a retrieved set is the measure of its relevance to the query that generated the result set.

Effectiveness of an IR system

- The effectiveness of an IR system (i.e., the quality of its search results) is normally determined by:
- **Precision:** What fraction of the returned results are relevant to the information need?
- **Recall:** What fraction of the relevant documents in the collection were returned by the system?
- **What is the best balance between the two?**
- Easy to get perfect recall: just retrieve everything
- Easy to get good precision: retrieve only the most relevant